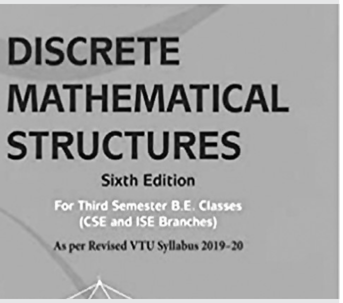

Source: https://www.amazon.in/Discrete-Mathematical-Structures-Dr-D-S-C/dp/9388478398

25MT205 DISCRETE MATHEMATICAL STRUCTURES

Hours Per Week :

L	T	P	SL	C
2	2	0	2	3

PREREQUISITE KNOWLEDGE: Basic understanding of set theory, basic algebra, and logic.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the fundamental concepts of discrete mathematics that are essential in computer science. It includes the study of algebraic structures, logic, graph theory, Boolean algebra, and combinatorics. The objective is to provide a solid foundation for students to understand theoretical concepts and apply them in computer science applications such as algorithm design, database theory, and artificial intelligence.

MODULE-1

UNIT-1 **6L+6T+0P+6SL=18 Hours**

ALGEBRAIC STRUCTURE

Binary operation, Relation, Semi Group, Monoid, Group Poset, Hasse Diagram, Lattice.

UNIT-2 **6L+6T+0P+6SL=18 Hours**

MATHEMATICAL LOGIC

Statements and notation, Connectives, Well-formed formulas, Truth table, Tautology, Equivalence Implication, Quantifiers, Predicate Calculus, Inference theory, Logic Gates operations.

PRACTICES:

- Constructing Hasse diagrams for given Posets
- Solving problems on logical equivalences
- Creating truth tables for compound propositions
- Applying predicate logic to simple problems
- Implementing basic logic gates in simulations
- Verifying tautologies using truth tables

MODULE-2

UNIT-1 **6L+6T+0P+6SL=18 Hours**

NORMAL FORMS

CNF, DNF, PDNF, PCNF, Conversion of CNF to DNF and vice versa. Minimization of Boolean Function (Karnaugh Maps).

UNIT-2 **6L+6T+0P+6SL=18 Hours**

GRAPH THEORY

Definition of graph, Incidence, Degree, Walk, Path, Circuit, Special types of graphs, Connected graph, Weighted graph, Matrix representation of graph, Graph Isomorphism, Euler and Hamiltonian path and circuit, Eulerian graph, Hamiltonian graph, Planar graphs, Bipartite graph, Tree, Spanning Trees, Binary Trees, Decision Trees, Applications in AI.

UNIT-3 **6L+6T+0P+6SL=18 Hours**

GRAPH ALGORITHMS

Shortest path algorithms (Dijkstra's algorithm and Floyd-Warshall's algorithm), DFS, BFS algorithm.

Minimal spanning tree (Kruskal's algorithm and Prim's algorithm), Graph coloring, Chromatic number, The four color problem.

PRACTICES:

- Graph traversal exercises (DFS, BFS)
- Karnaugh Map simplifications for Boolean expressions
- Implementing Dijkstra's algorithm in Python/C++
- Drawing and analyzing binary trees and decision trees
- Converting between CNF and DNF
- Designing minimal spanning trees using Kruskal's and Prim's algorithms

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Understand fundamental concepts of algebraic structures, relations and mathematical logic, including propositional and predicate calculus.	Analyze	1,11
2	Apply Boolean algebra techniques such as normal forms (CNF, DNF) and Karnaugh Maps for function minimization and logical simplifications.	Apply	1,3,5,
3	Apply combinatorial techniques, including permutations, combinations, recurrence relations and generating functions, to solve discrete mathematics problems.	Analyze	1,2,5,
4	Analyze different graph structures, spanning trees and their properties, including Eulerian and Hamiltonian circuits, isomorphisms and planarity.	Analyze	2,3,4
5	Evaluate graph algorithms and minimum spanning trees to optimize computational problems.	Evaluate	3,5,6,10,

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS):

4 QUALITY EDUCATION

8 DECENT WORK AND ECONOMIC GROWTH

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

11 SUSTAINABLE CITIES AND COMMUNITIES

12 RESPONSIBLE CONSUMPTION AND PRODUCTION

13 CLIMATE ACTION

17 PARTNERSHIPS FOR THE GOALS

INDIAN KNOWLEDGE SYSTEM (IKS)

SKILLS:

- ✓ *Familiarity of concepts of statements, logic and truth tables.*
- ✓ *Analyze closed form of discrete numeric function.*
- ✓ *Know some basic properties of graphs, trees and related discrete structures.*

TEXT BOOKS:

1. Kenneth H. Rosen, “Discrete Mathematics and Its Applications”. McGraw-Hill, 7th Edition, 2012.
2. J.P. Tremblay and R. Manohar, “Discrete Mathematical Structures with Applications to Computer Science”. Tata McGraw-Hill, 1st Edition, 2001.

REFERENCE BOOKS:

1. C.L. Liu and D.P. Mohapatra, “Elements of Discrete Mathematics”. McGraw-Hill, 4th Edition, 2010.
2. Bernard Kolman, Robert C. Busby, and Sharon Cutler Ross, “Discrete Mathematical Structures”. Pearson, 6th Edition, 2008.
3. Susanna S. Epp, “Discrete Mathematics with Applications”. Cengage Learning, 4th Edition, 2011.